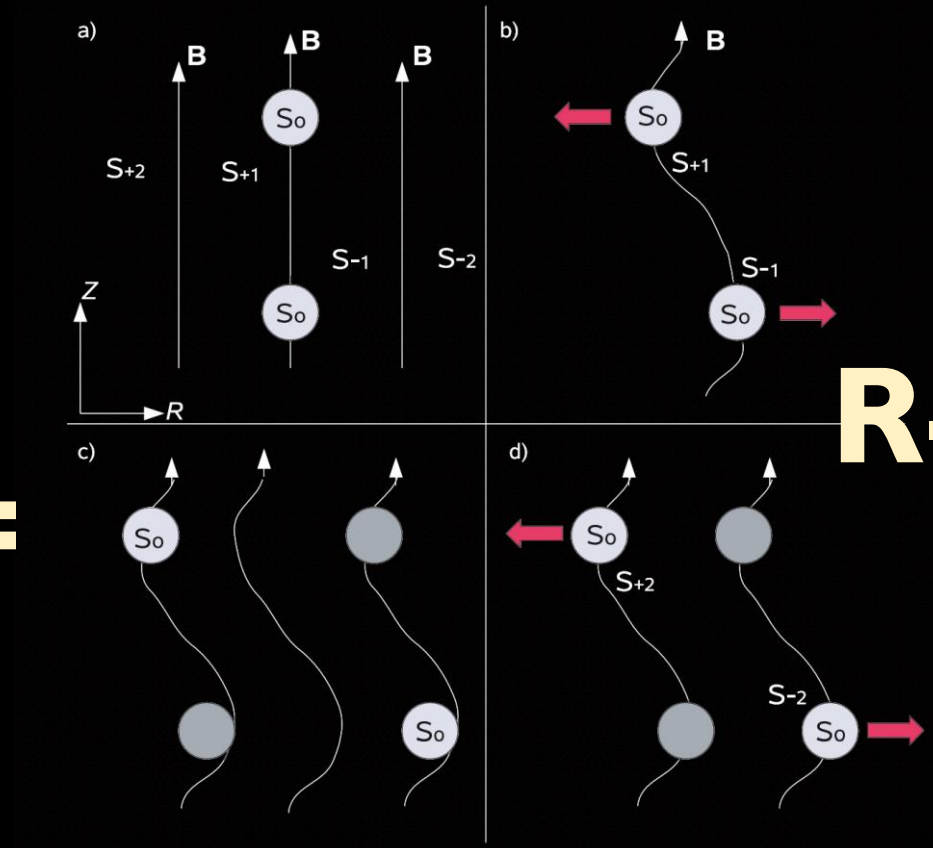


Can radial convection happen in protoplanetary discs even *with magnetic fields*?

COS
+ *B-field* =



R-DDI

Image credit:
Latter+ 2010

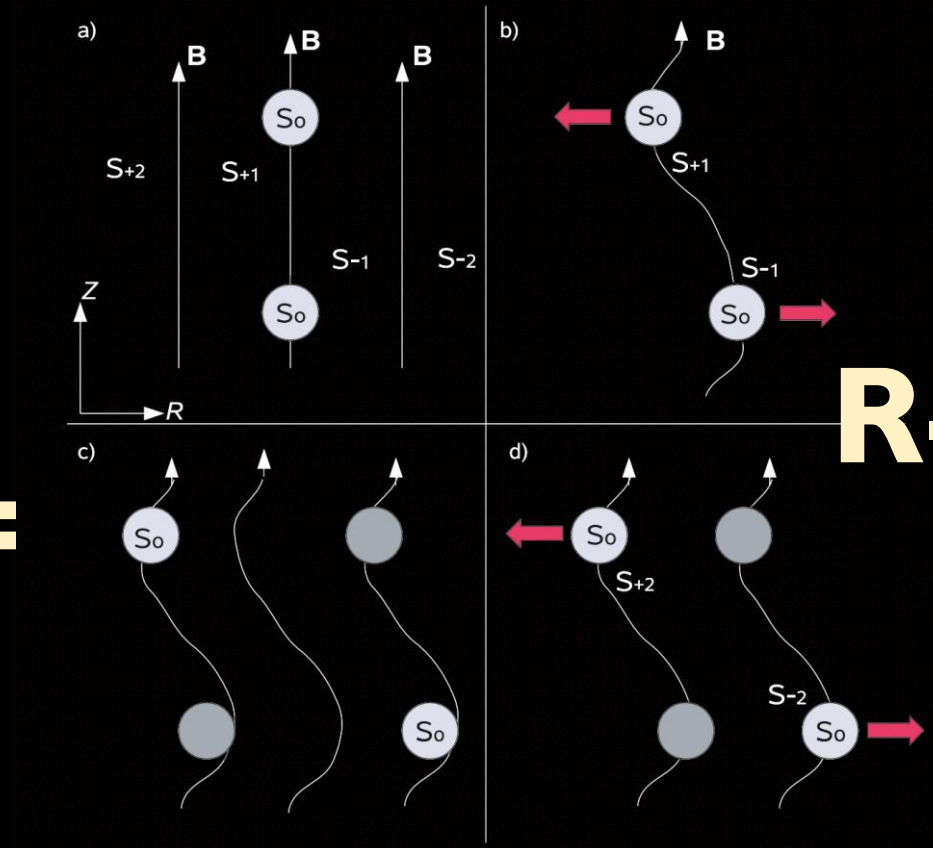
Michael Hammer

Max Planck Institute for Solar System Research (MPS)

Collaborator: Min-Kai Lin (ASIAA)

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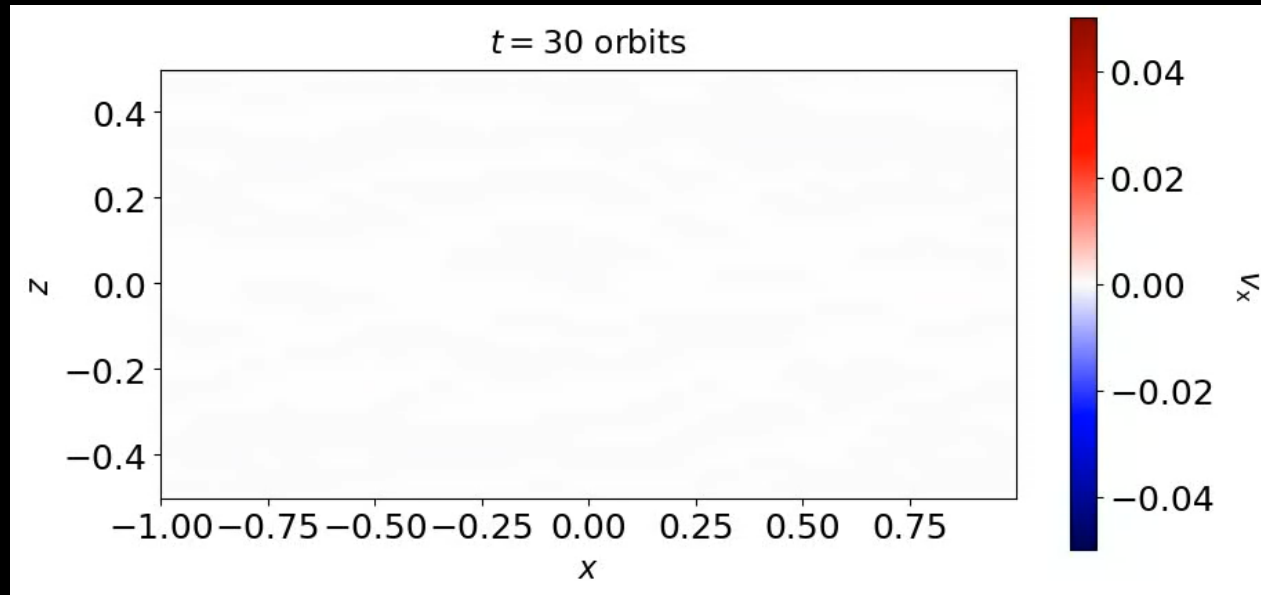
Max Planck Institute for Solar System Research (MPS)

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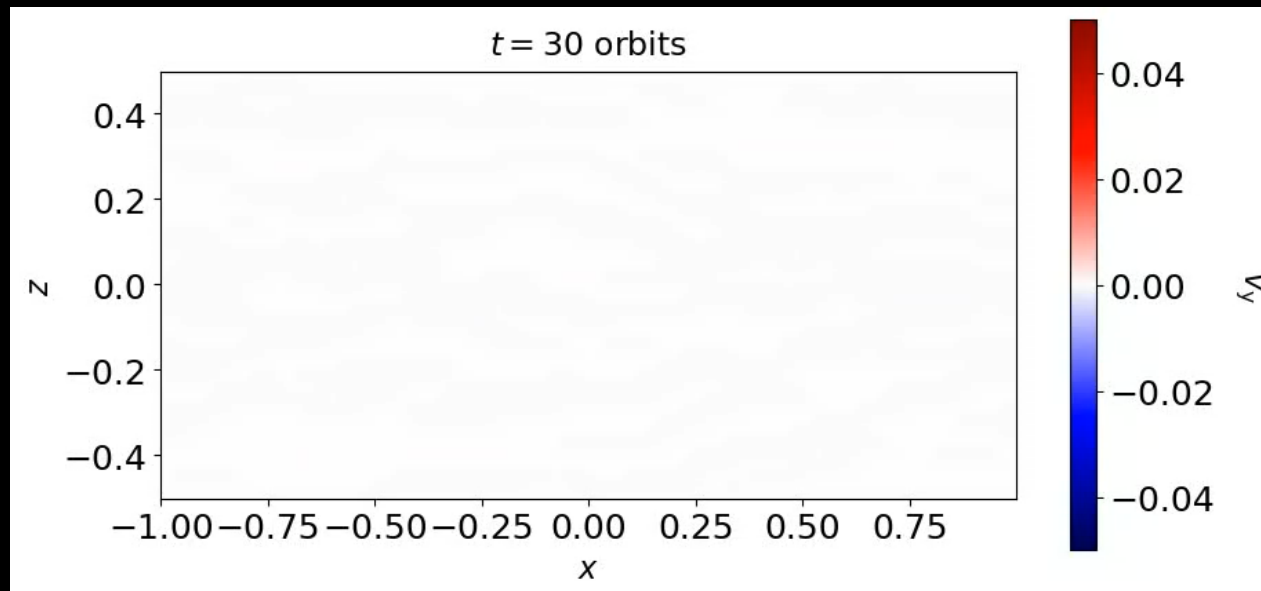
Can radial convection happen
in protoplanetary discs
even with magnetic fields?

Yes!

Convection w/ Magnetic Fields



Radial
velocity



Azimuthal
velocity

MH + M.-K. Lin, 2026,
in prep.

Is there convection
in protoplanetary discs?

Not exactly!

(It is suppressed by the disc's rotation.)

**Instead, PPDs can have the
convective overstability (COS)!**

(It's like convection, but the parcels move in epicycles.)

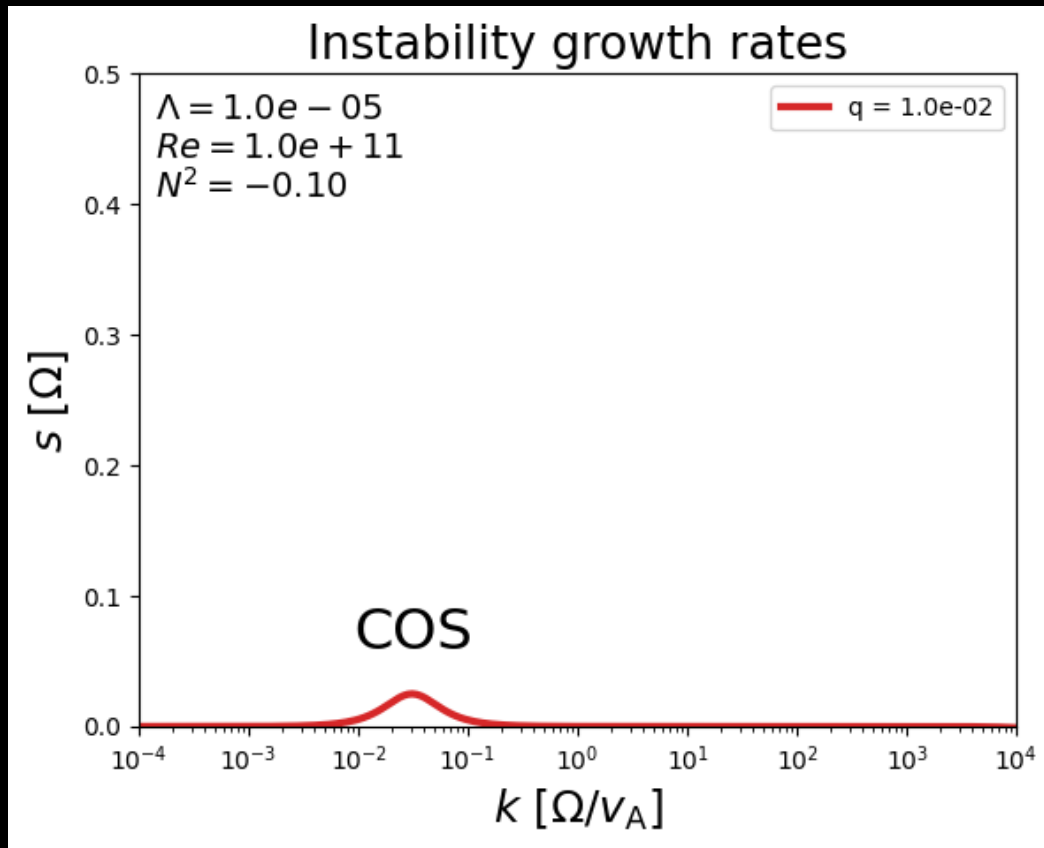
Is there convective overstability (COS) in protoplanetary discs?

Possibly!

but it can be suppressed by...

Magnetic fields!

Convective overstability (COS) w/ weak magnetic fields



Elsasser number

$$\Lambda = \frac{v_A^2}{\eta \Omega}$$

Magnetic field
strength

Magnetic
diffusion

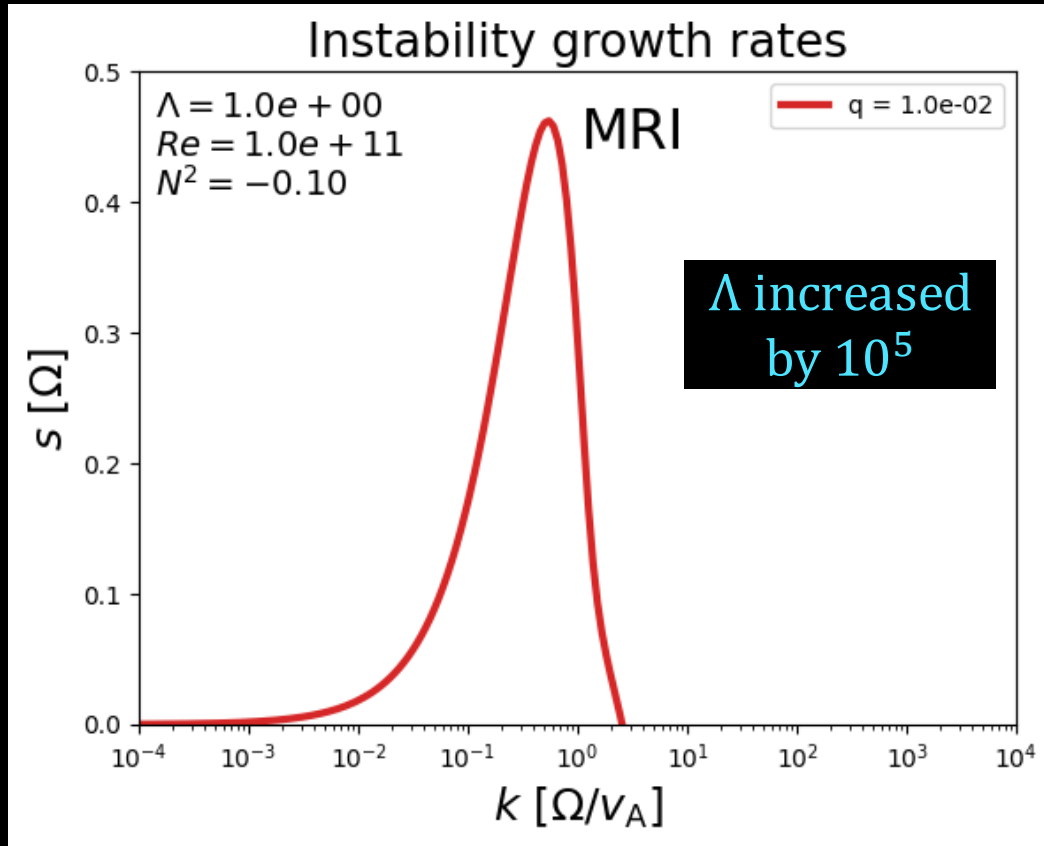
Roberts number

$$q = \frac{\xi}{\eta}$$

Thermal
diffusion

Magnetic
diffusion

Convective overstability (COS) w/ strong magnetic fields



Elsasser number

$$\Lambda = \frac{v_A^2}{\eta \Omega}$$

Magnetic field
strength

Magnetic
diffusion

Roberts number

$$q = \frac{\xi}{\eta}$$

Thermal
diffusion

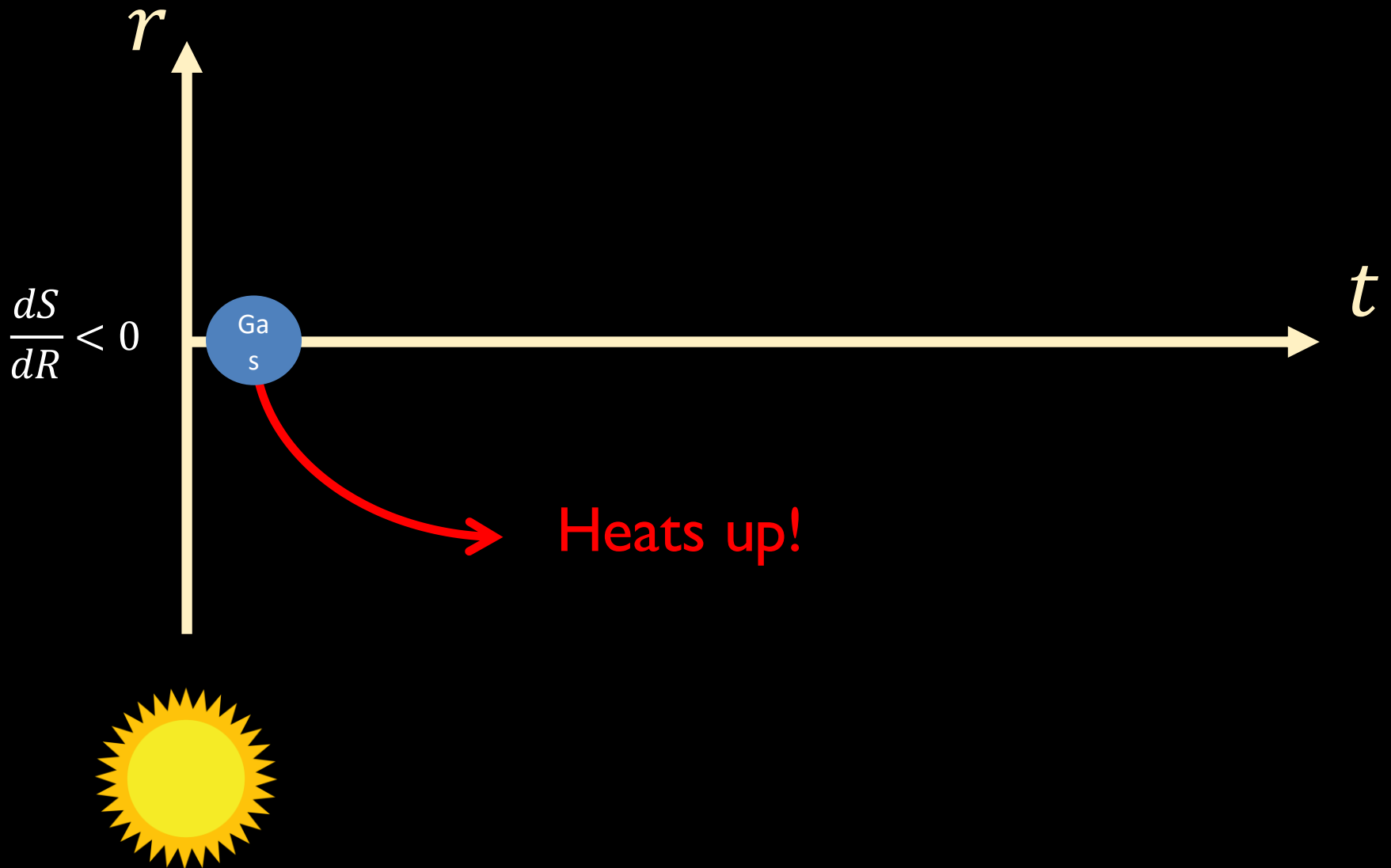
Magnetic
diffusion

COS replaced by MRI!

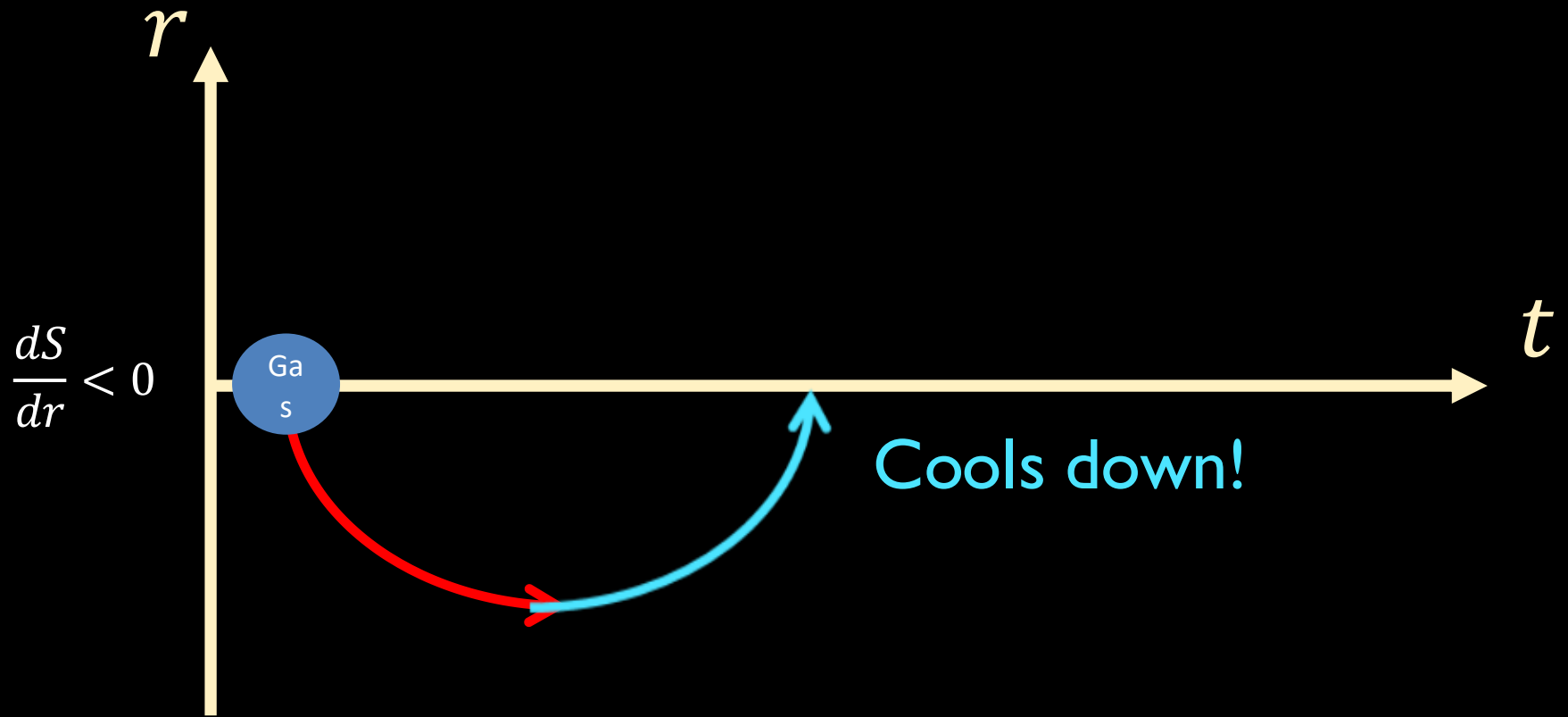
Can convection still
overcome the MRI?

Yes, with magnetic fields!

Convective Overstability (COS) mechanism

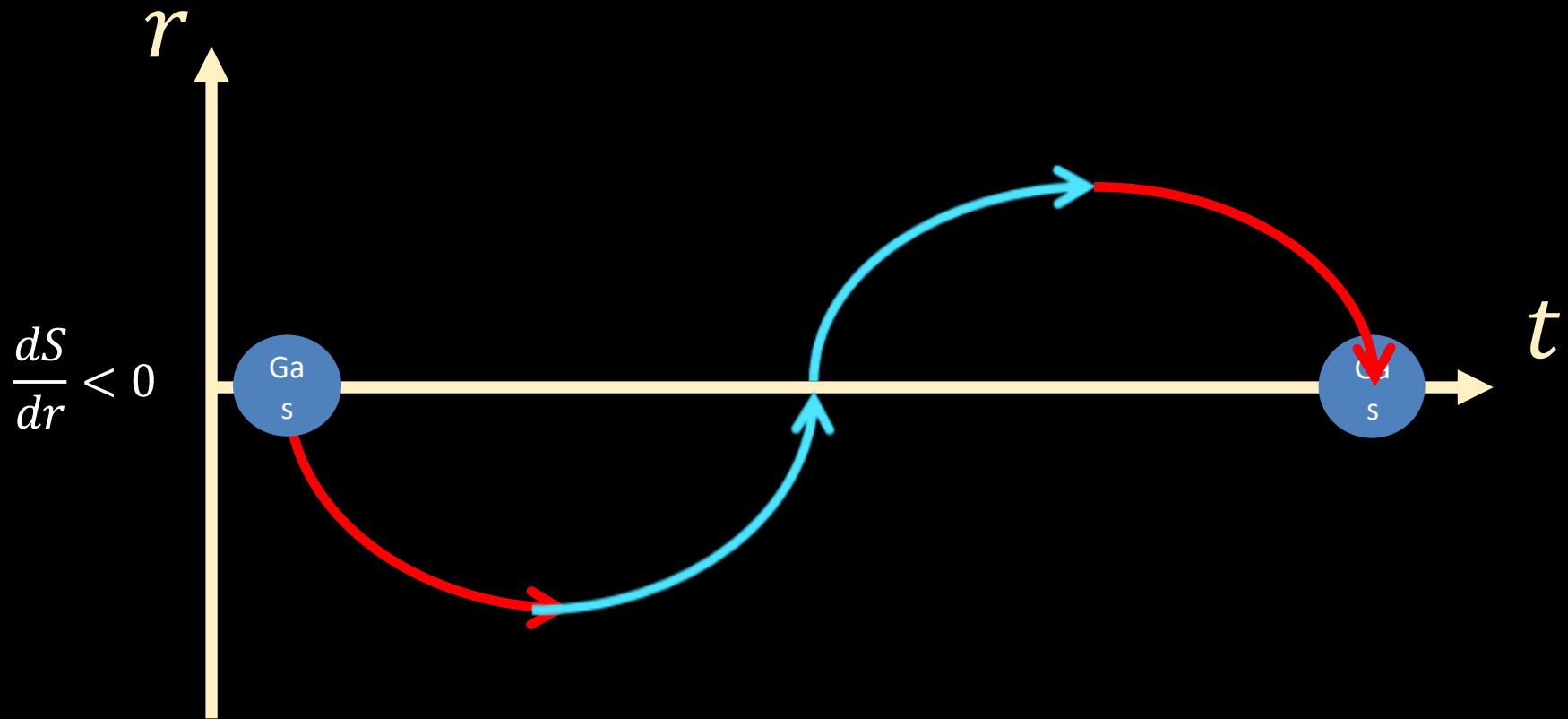


Convective Overstability (COS) mechanism



With immediate cooling,
this is stable. (No COS!)

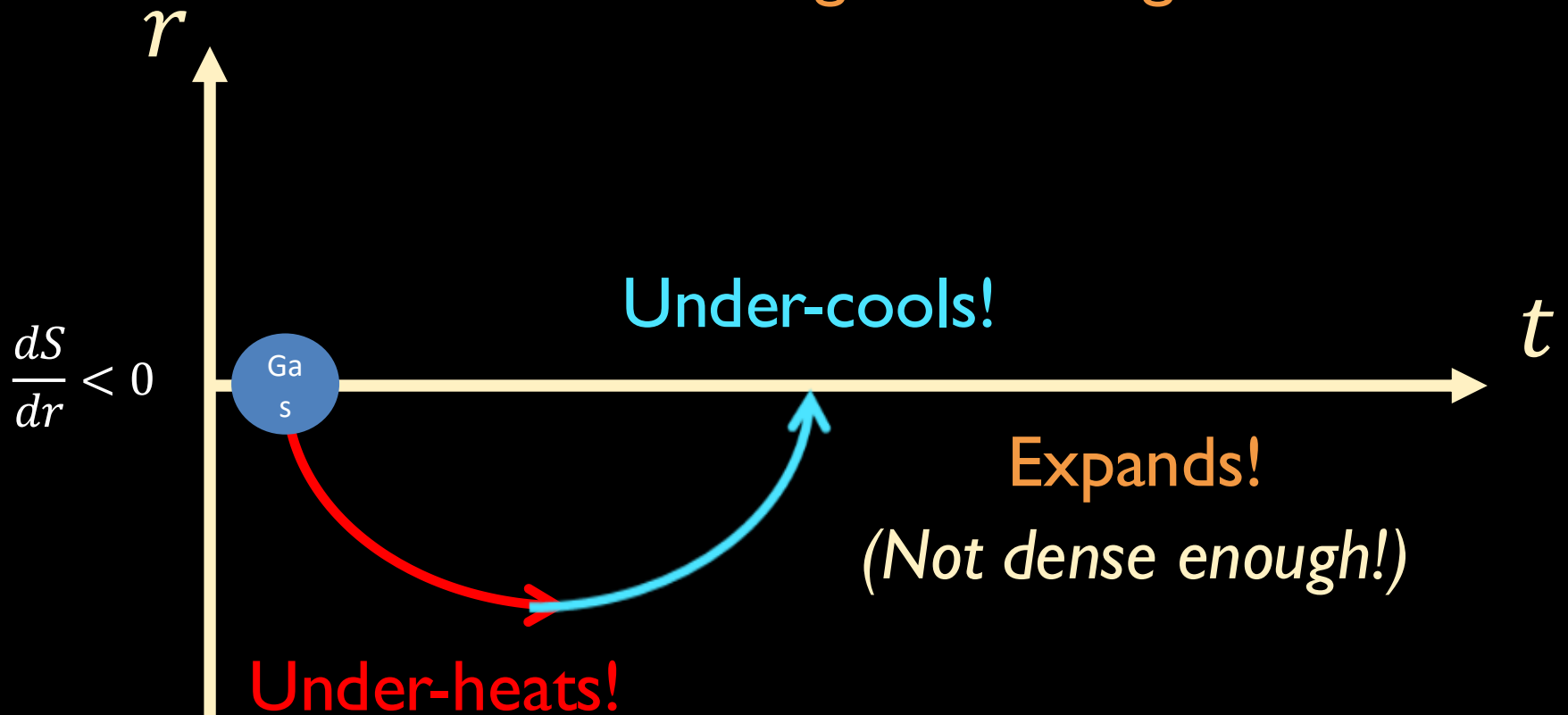
Convective Overstability (COS) mechanism



With immediate cooling,
this is stable. (No COS!)

Convective Overstability (COS) mechanism

Pressure changes don't lag behind!



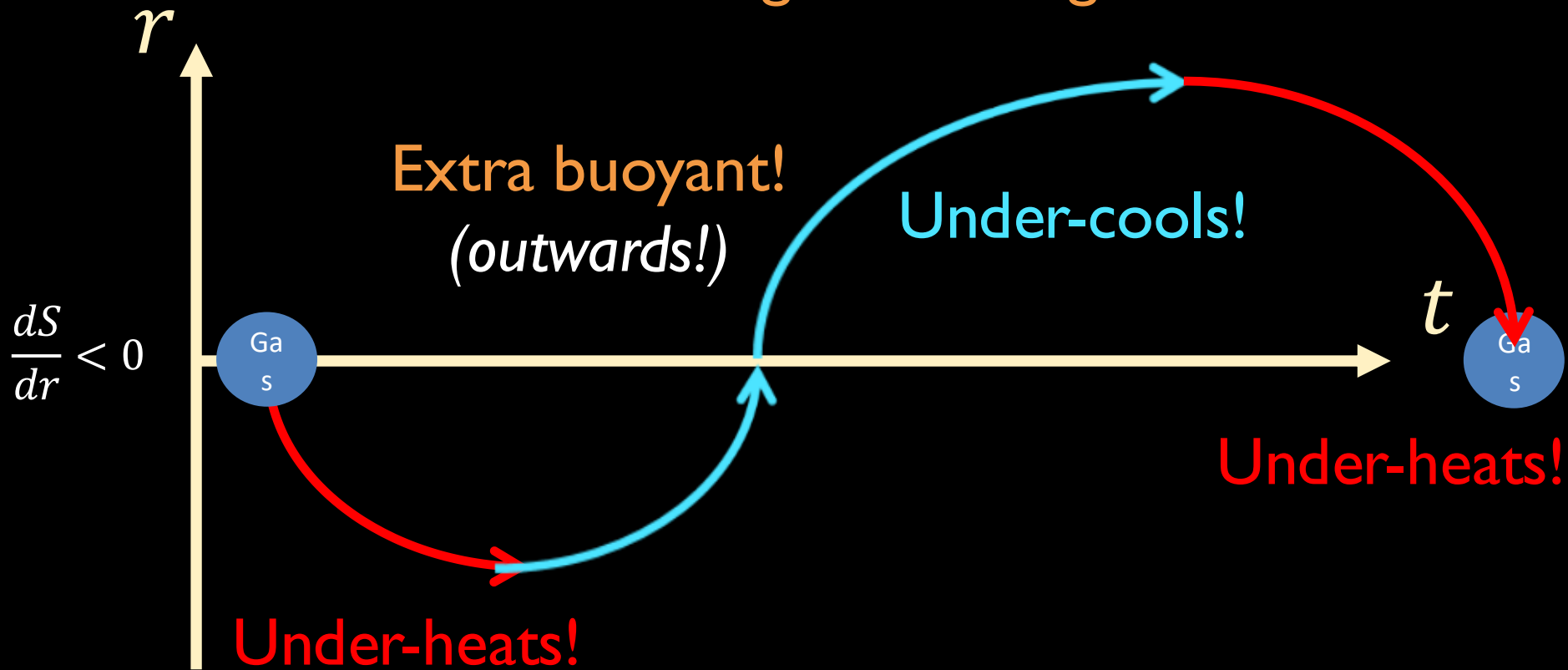
But thermal diffusion
(i.e. cooling) takes time!

$$\xi \approx 1$$

Thermal
diffusion

Convective Overstability (COS) mechanism

Pressure changes don't lag behind!



But thermal diffusion
(i.e. cooling) takes time!

$$\xi \approx 1$$

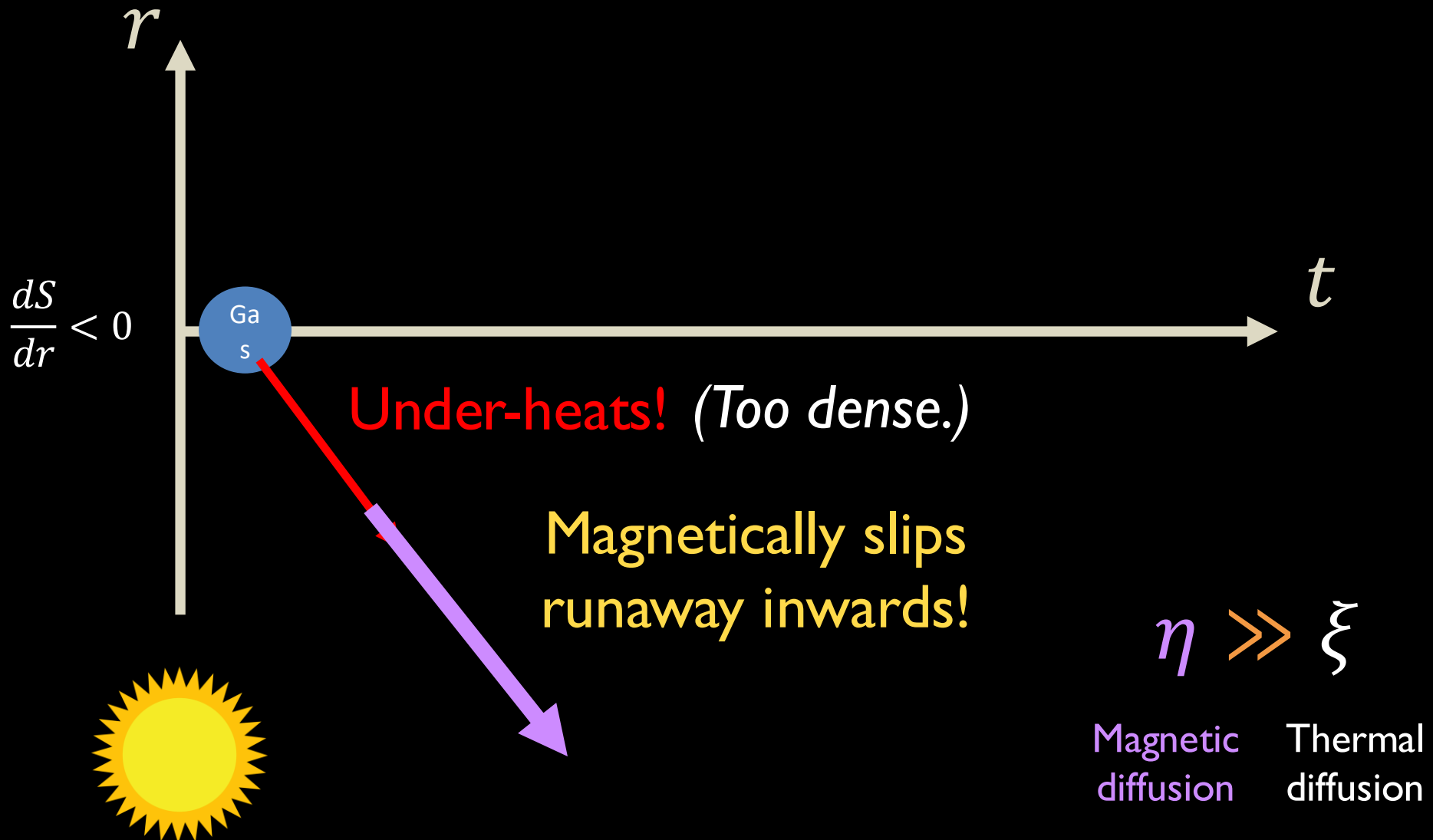
Thermal
diffusion

How can we make buoyancy
overcome the MRI?

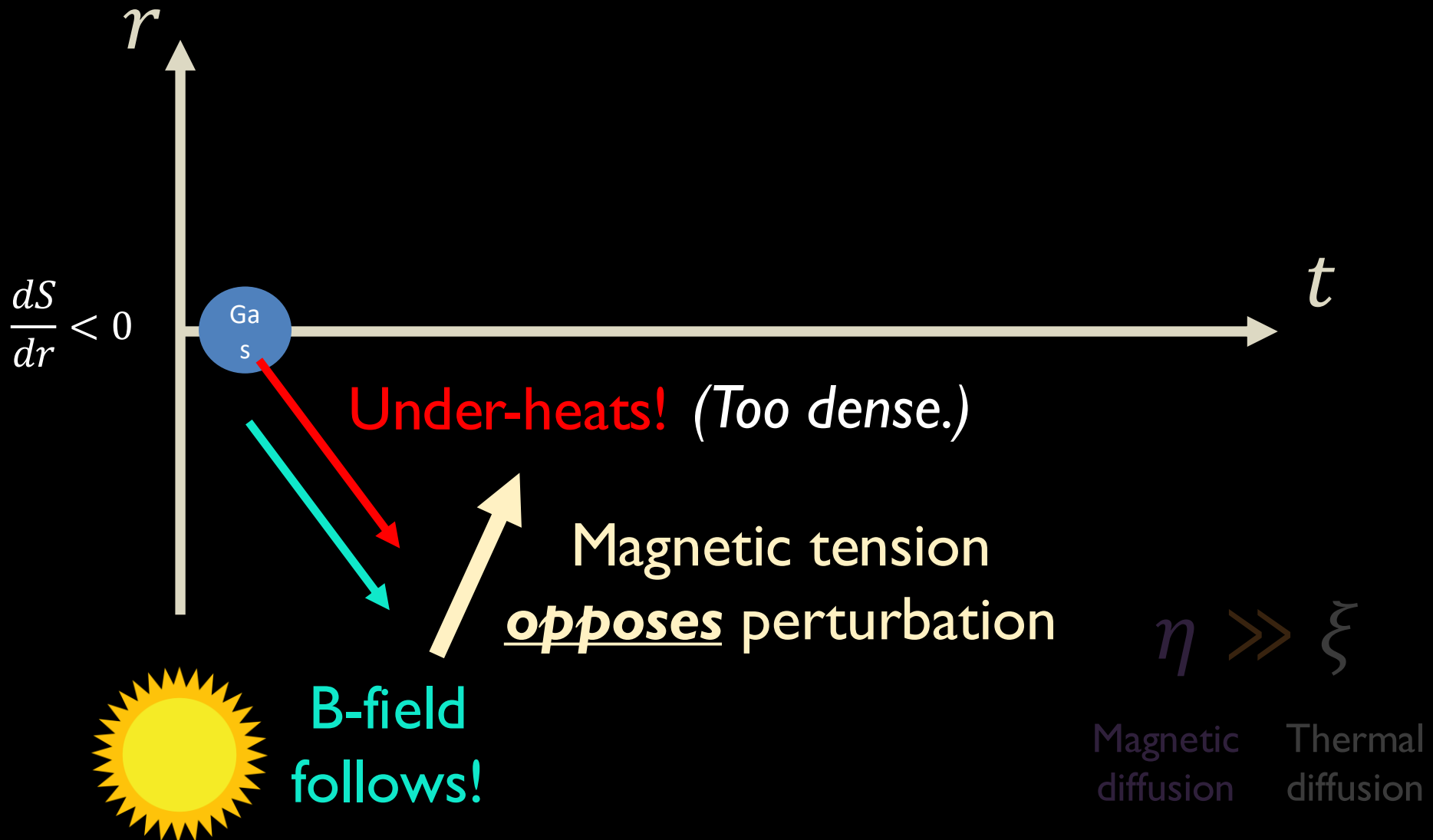
Add in magnetic slippage!

(through magnetic diffusion)

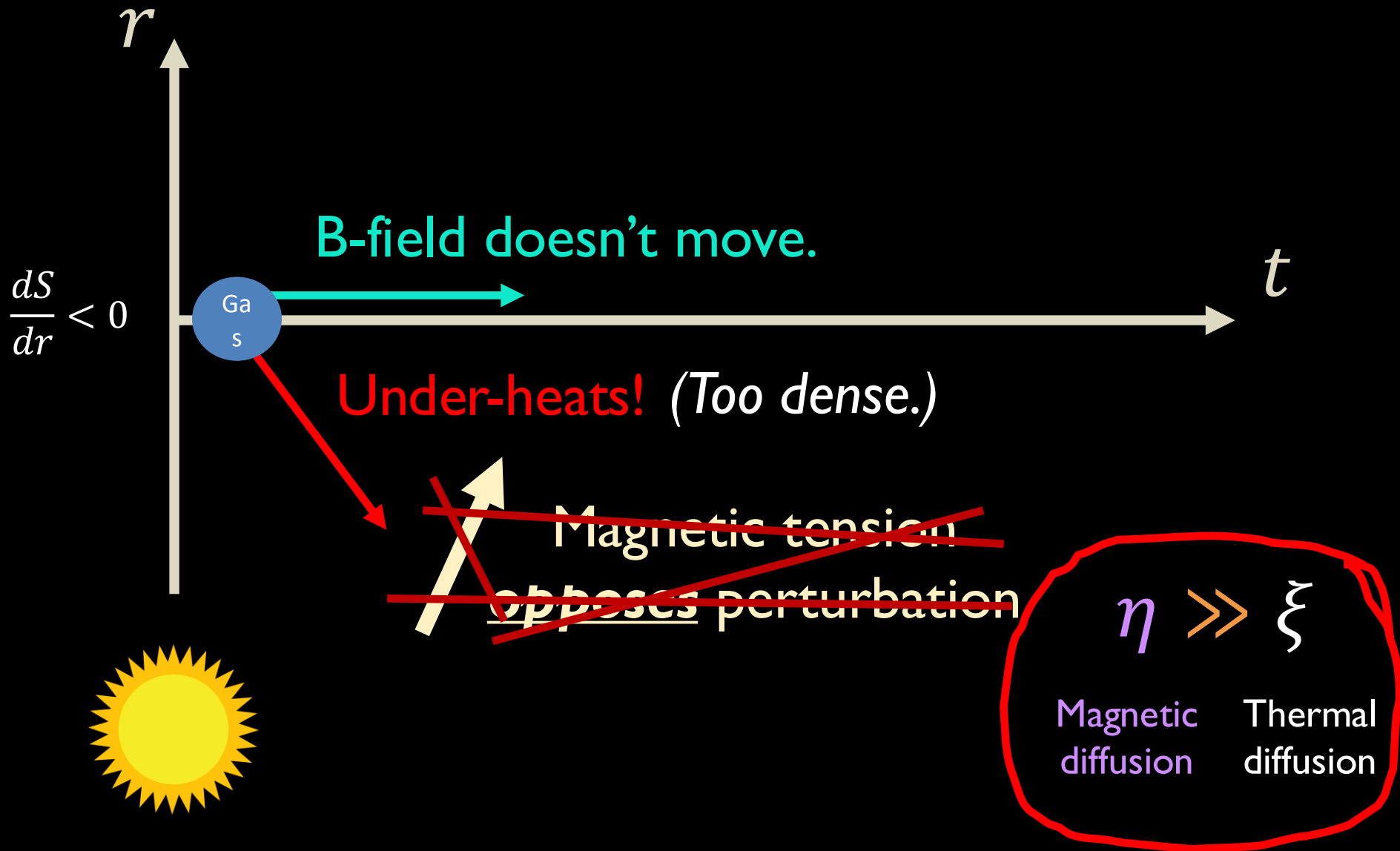
Add in *magnetic fields!*



Add in *magnetic fields!*



Add in *magnetic fields*!

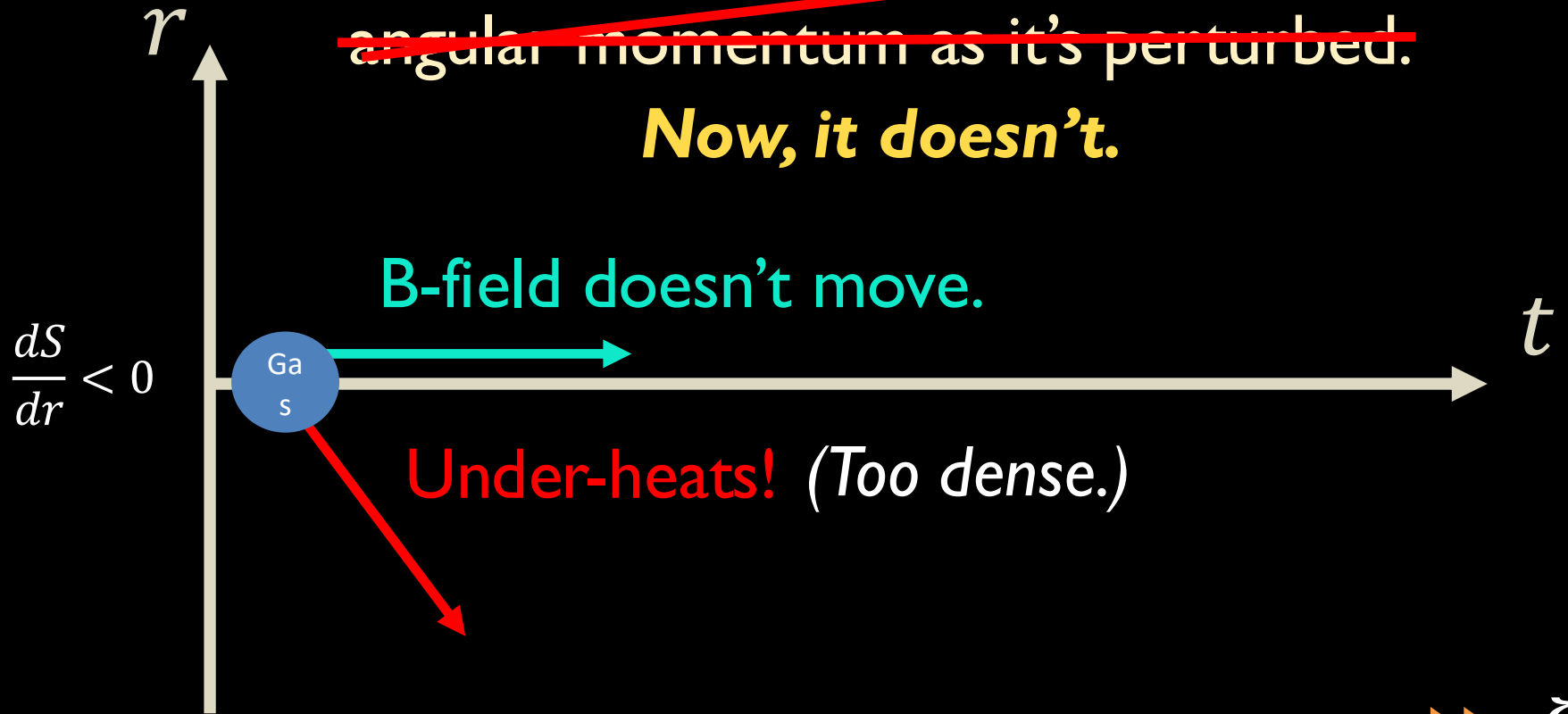


Resistive Double-Diffusive Instability (R-DDI)

Latter+ 2010

~~Normally, the gas needs to conserve angular momentum as it's perturbed.~~

Now, it doesn't.



$$\eta \gg \xi$$

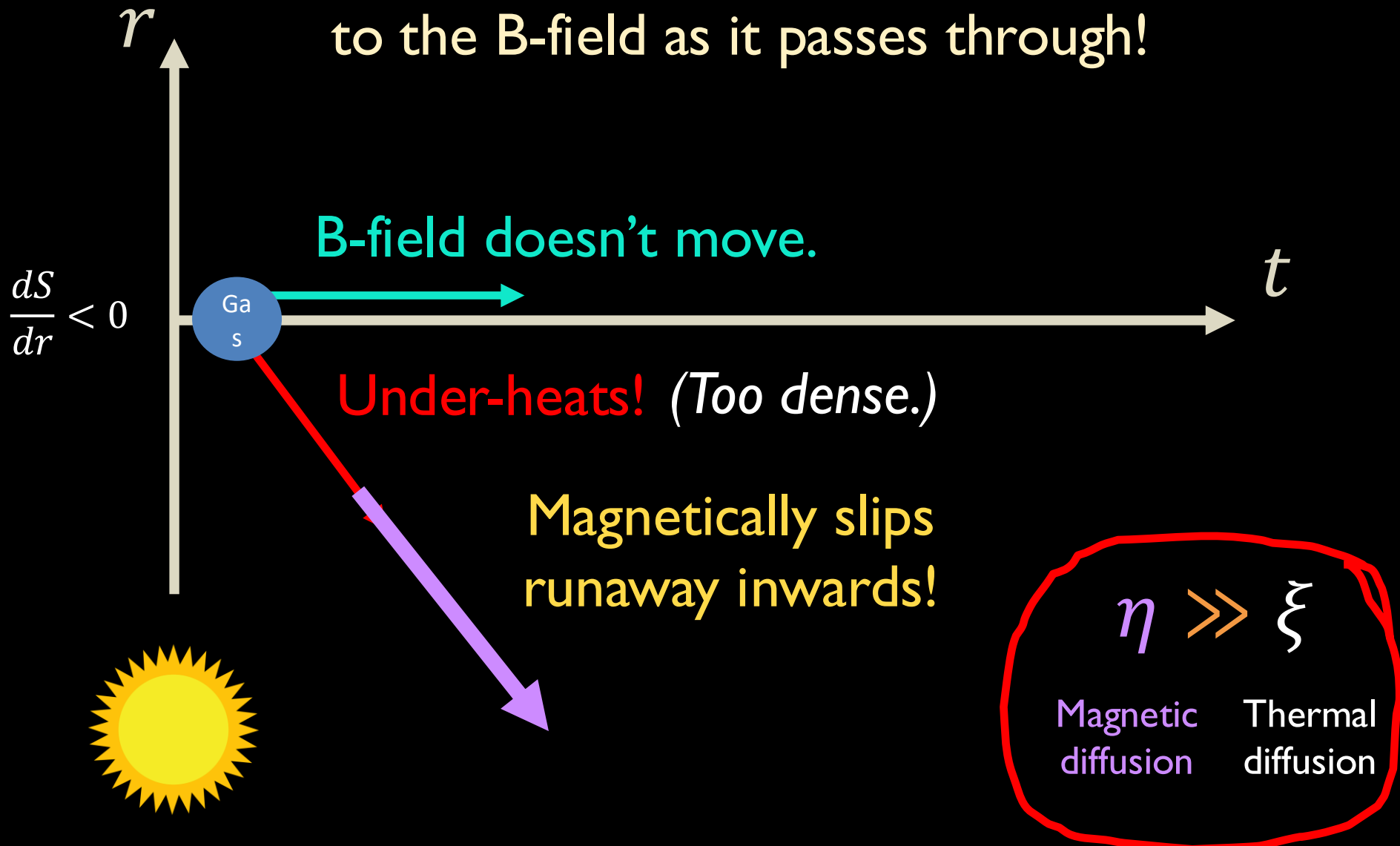
Magnetic
diffusion

Thermal
diffusion

Resistive Double-Diffusive Instability (R-DDI)

Latter+ 2010

The gas sheds angular momentum to the B-field as it passes through!



Convective Overstability (COS)

Resistive Double- Diffusive Instability (R-DDI)

Latter+ 2010

Both require $\frac{dS}{dr} < 0$.

(small $\frac{d \ln P}{d \ln r} < 0$, such as near a pressure bump)

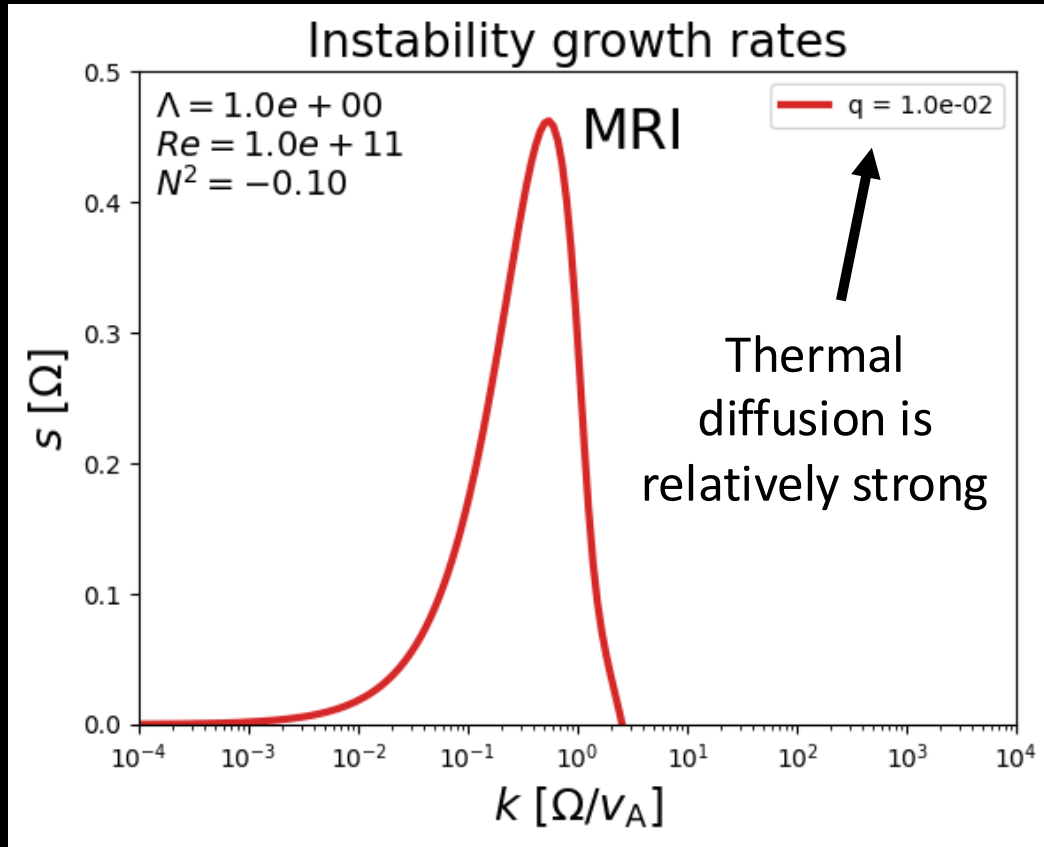
The **buoyancy force** accelerates the perturbation instead of opposing it.

Instabilities caused by not having
fast enough cooling to keep up w/ the pressure change.

Made possible by
thermal lag
during epicycles.

Made possible by
magnetic slippage from
diffusion (*Ohmic resistivity*).

Convective overstability (COS) w/ strong magnetic fields



Elsasser number

$$\Lambda = \frac{v_A^2}{\eta \Omega}$$

Magnetic field
strength

Magnetic
diffusion

Roberts number

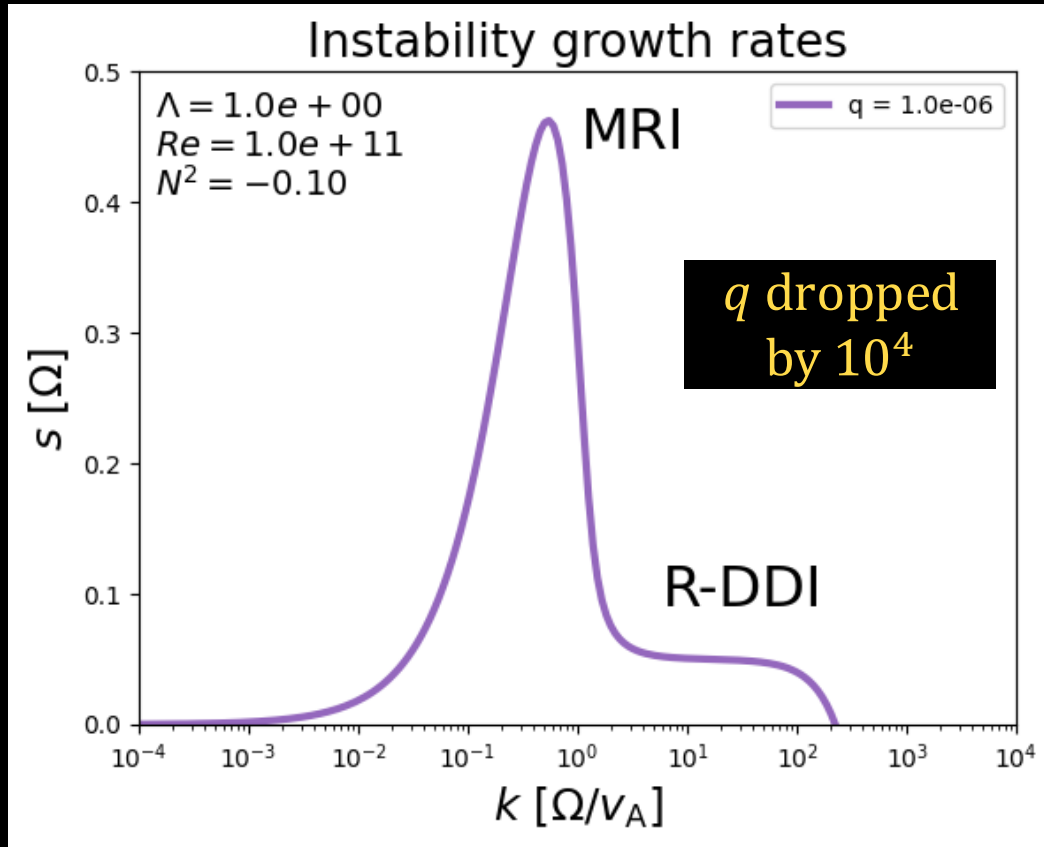
$$q = \frac{\xi}{\eta}$$

Thermal
diffusion

Magnetic
diffusion

COS replaced by MRI!

MRI w/ strong resistivity (Resistive Double-Diffusive Instability)



Elsasser number

$$\Lambda = \frac{v_A^2}{\eta \Omega}$$

Magnetic field
strength

Magnetic
diffusion

Roberts number

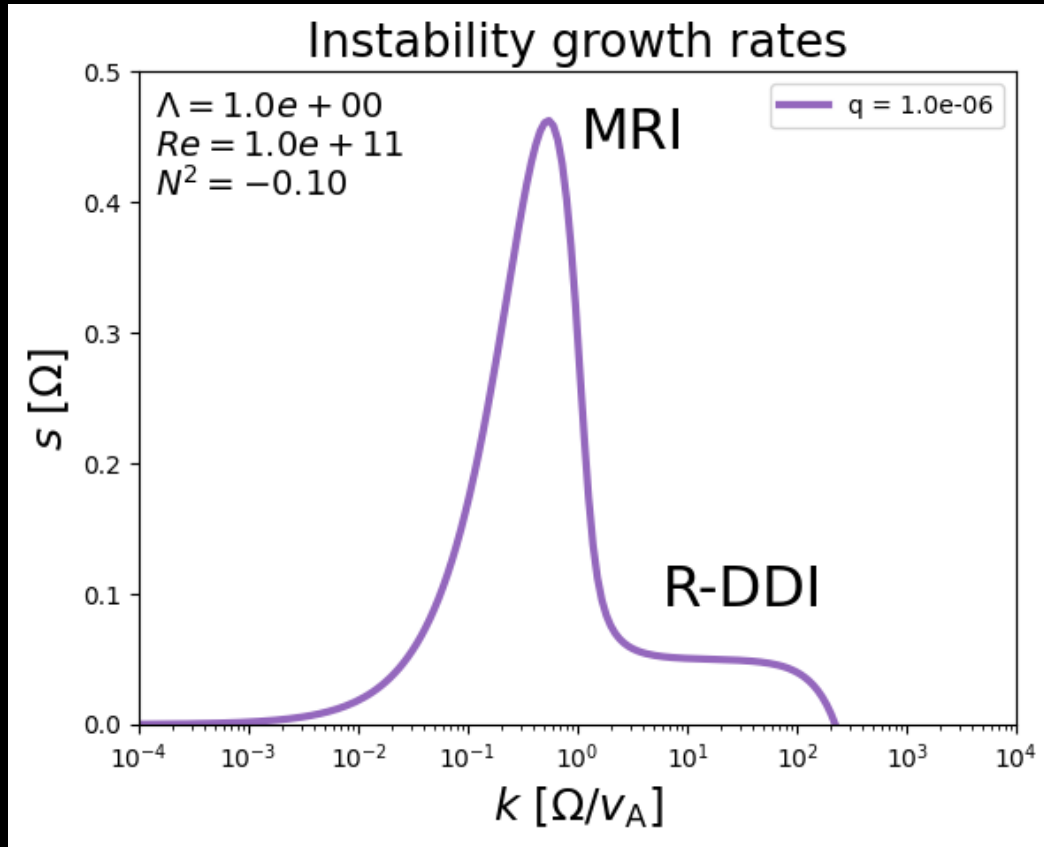
$$q = \frac{\xi}{\eta}$$

Thermal
diffusion

Magnetic
diffusion

MRI accompanied by R-DDI!

MRI w/ strong resistivity (Resistive Double-Diffusive Instability)



Elsasser number

$$\Lambda = \frac{v_A^2}{\eta \Omega}$$

Magnetic field
strength

Magnetic
diffusion

Roberts number

$$q = \frac{\xi}{\eta}$$

Thermal
diffusion

Magnetic
diffusion

But MRI still grows faster!

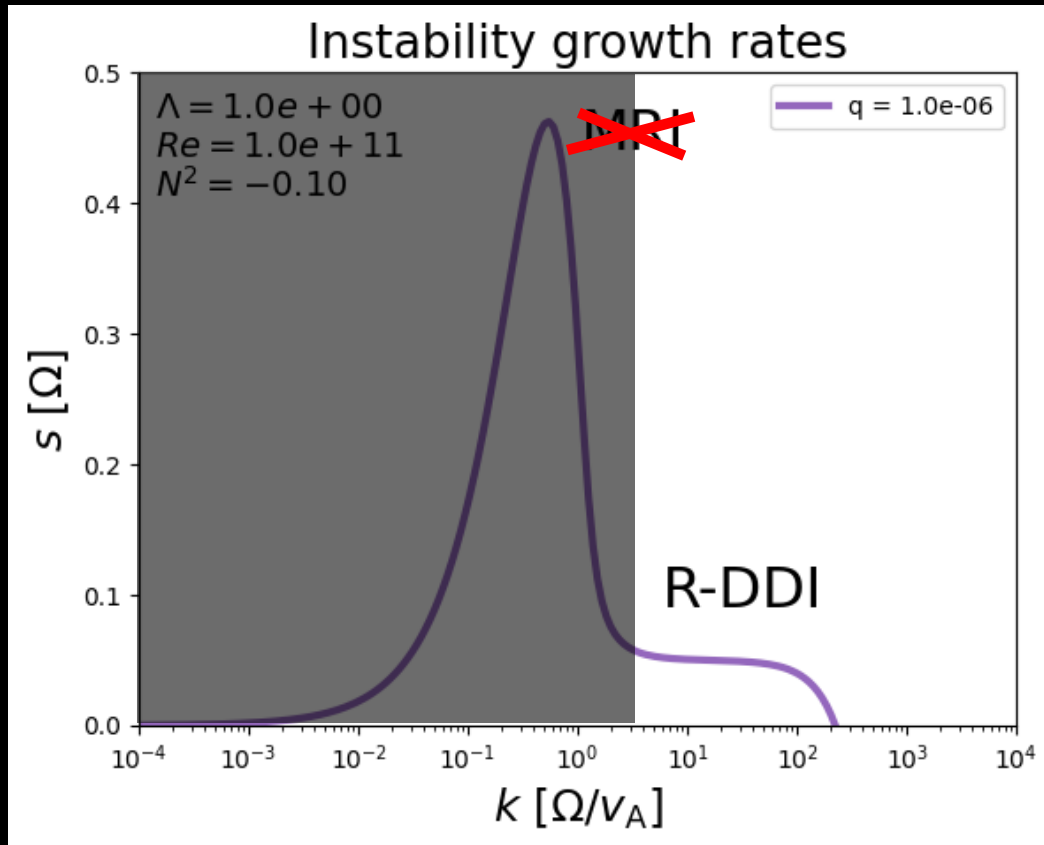
Can R-DDI overcome MRI?

Yes!

We don't expect MRI to
operate so easily.

It has two main weaknesses.

At what parameters does R-DDI happen?



Elsasser number

$$\Lambda = \frac{v_A^2}{\eta \Omega}$$

Magnetic field
strength

Magnetic
diffusion

Roberts number

$$q = \frac{\xi}{\eta}$$

Thermal
diffusion

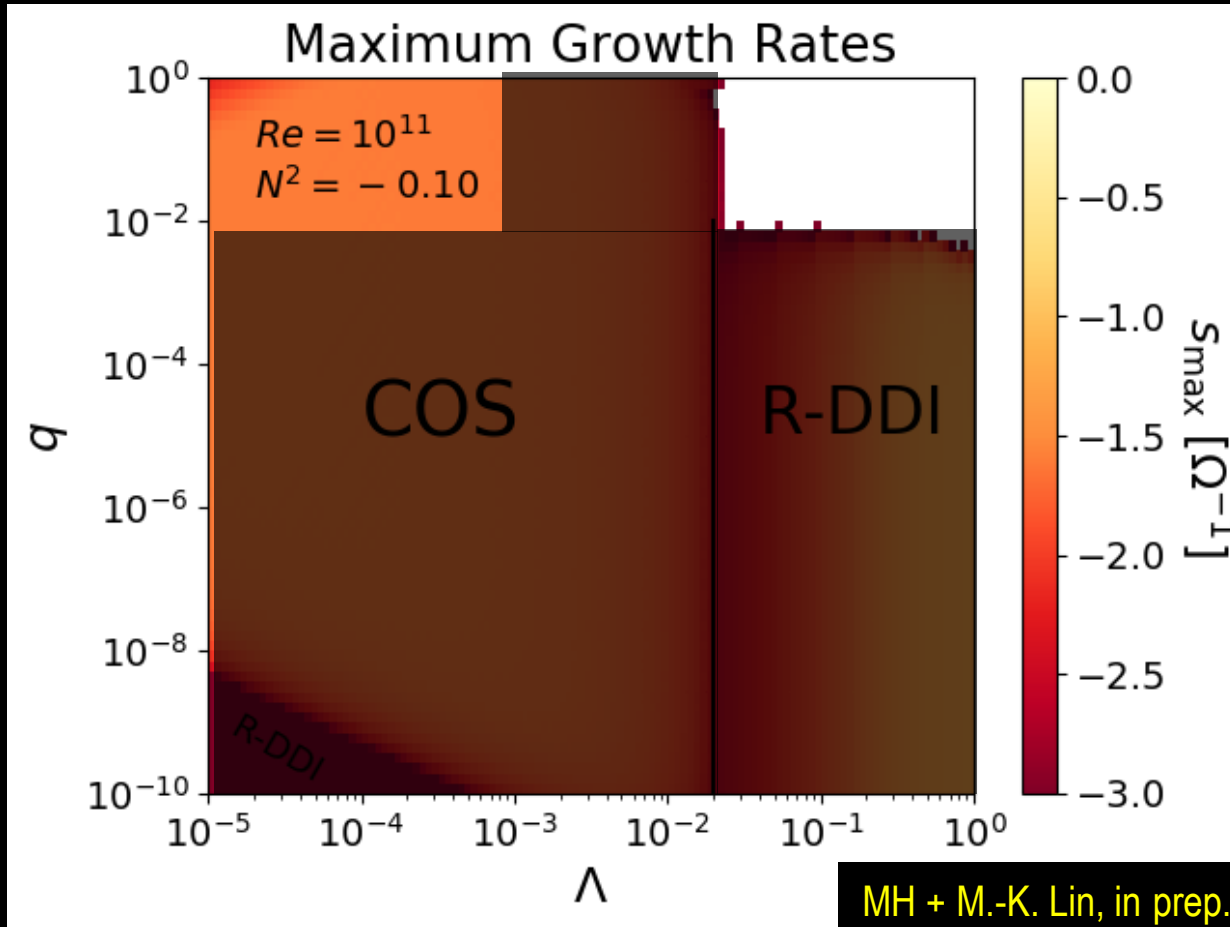
Magnetic
diffusion

$$\Lambda = 1$$

$$q = 10^{-6}$$

$$s_{max} \approx 0.05$$

At what parameters does R-DDI happen?



$$\Lambda = \frac{v_A^2}{\eta \Omega}$$

Magnetic field
strength

Magnetic
diffusion

$$q = \frac{\xi}{\eta}$$

Thermal
diffusion

Magnetic
diffusion

R-DDI needs

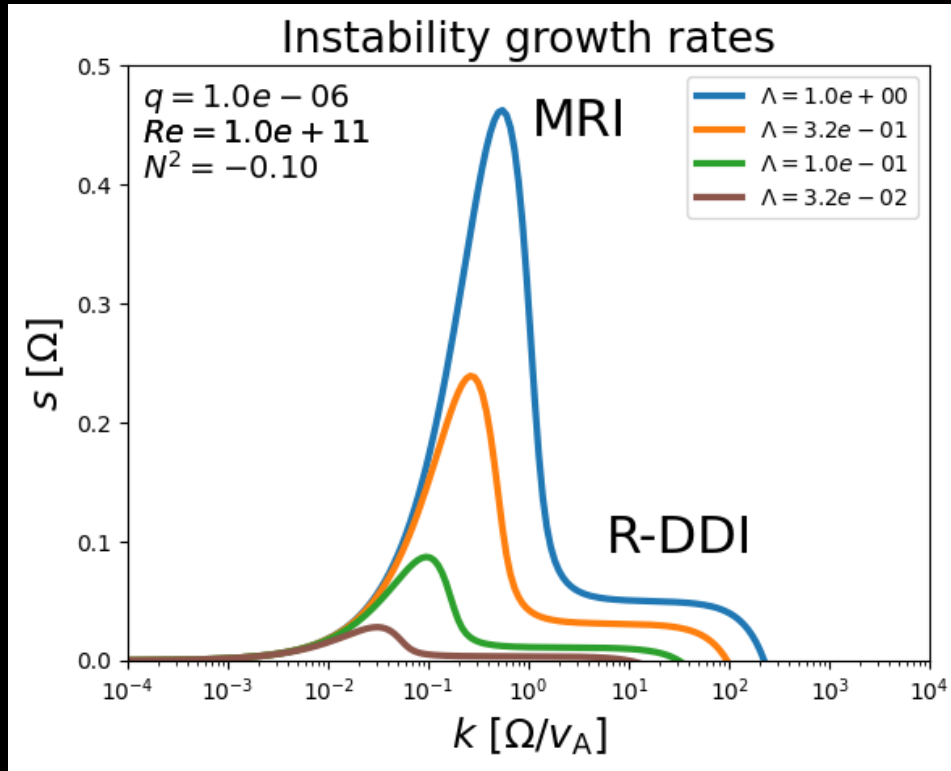
$$q < 10^{-2}$$

COS survives if
 $\Lambda \leq 10^{-2}$

R-DDI happens if
 $\Lambda > 10^{-2}$

*(Remember the
MRI is excluded!)*

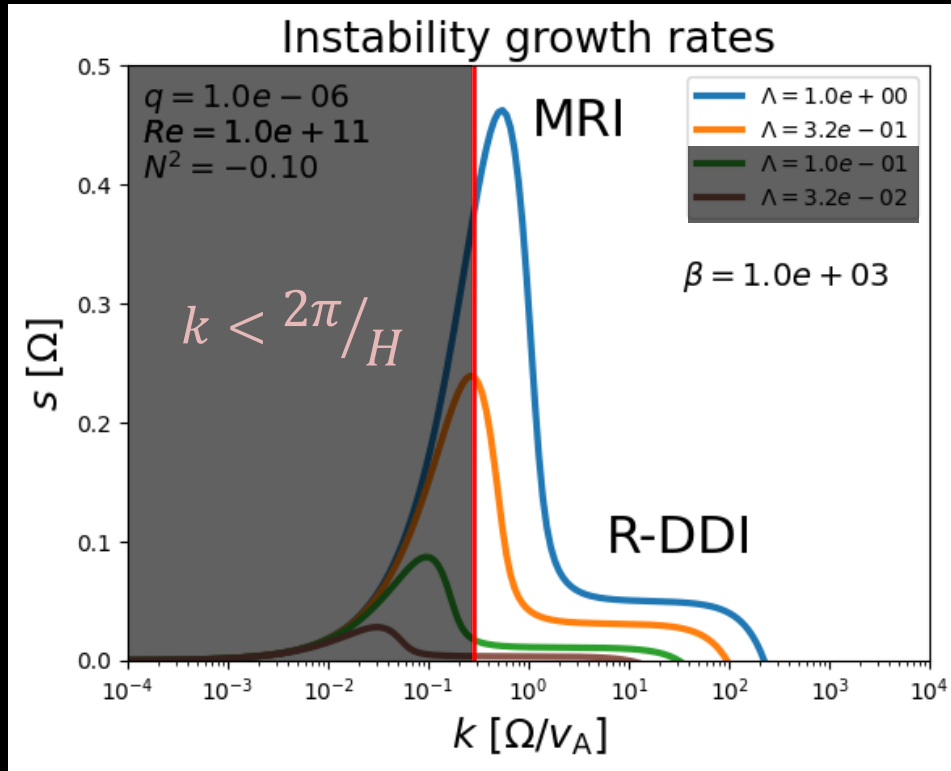
How to (*partially*) exclude the MRI



Lower Λ = Lower ionization rate
(*More non-ideal*)

MRI is weaker,
but still stronger than R-DDI!

How to (*partially*) exclude the MRI



Lower Λ = Lower ionization rate
(More non-ideal)

MRI is weaker,
but still stronger than R-DDI!

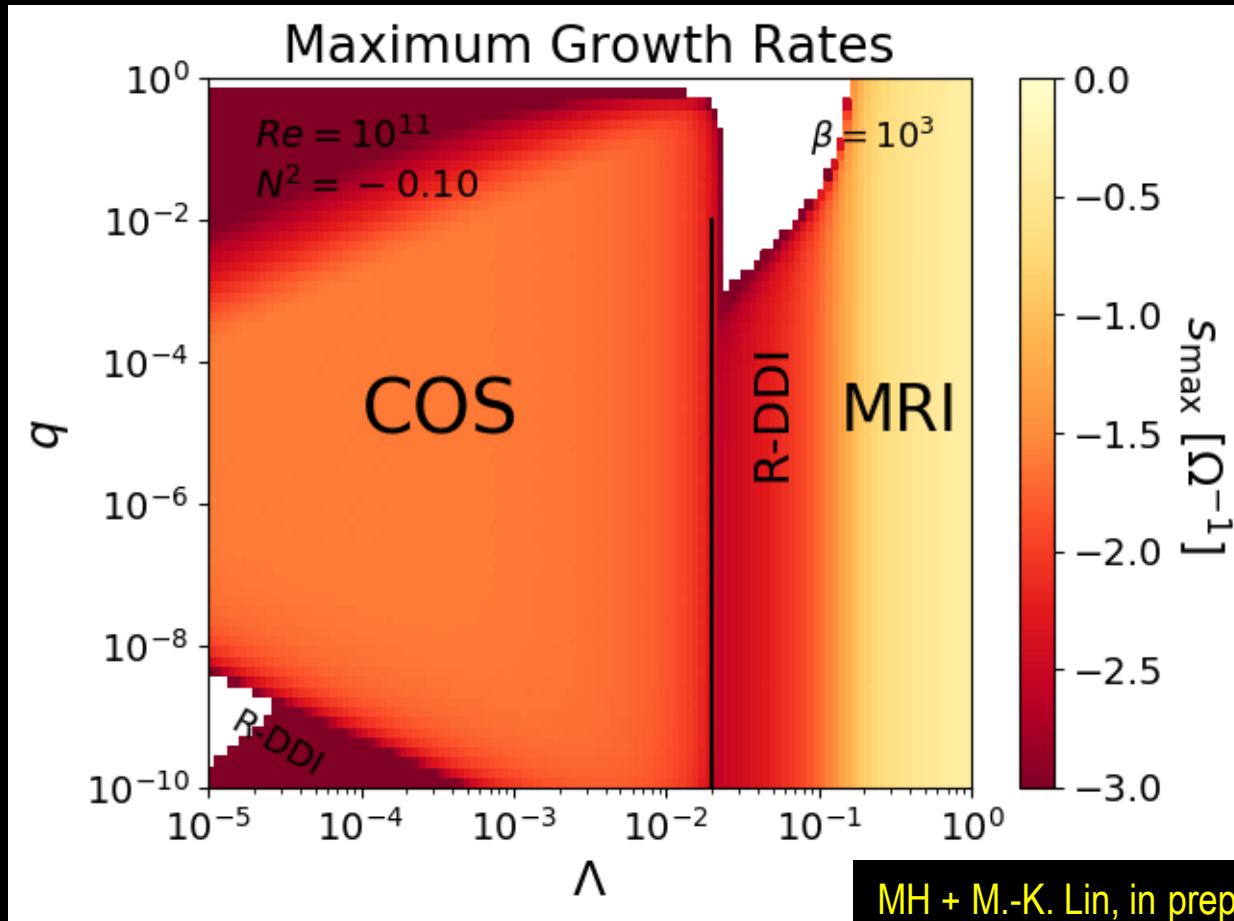
The scale k_{MRI}
becomes larger!
(k_{MRI} decreases)

MRI cannot operate if
 $k_{MRI} < 2\pi/H$



MRI excluded if
 $\Lambda \lesssim \sqrt{2} \beta^{-1/2}$

At what parameters does R-DDI happen?



$$\Lambda = \frac{v_A^2}{\eta \Omega}$$

Magnetic field strength

Magnetic diffusion

$$q = \frac{\xi}{\eta}$$

Thermal diffusion

Magnetic diffusion

R-DDI needs
 $q < 10^{-2}$

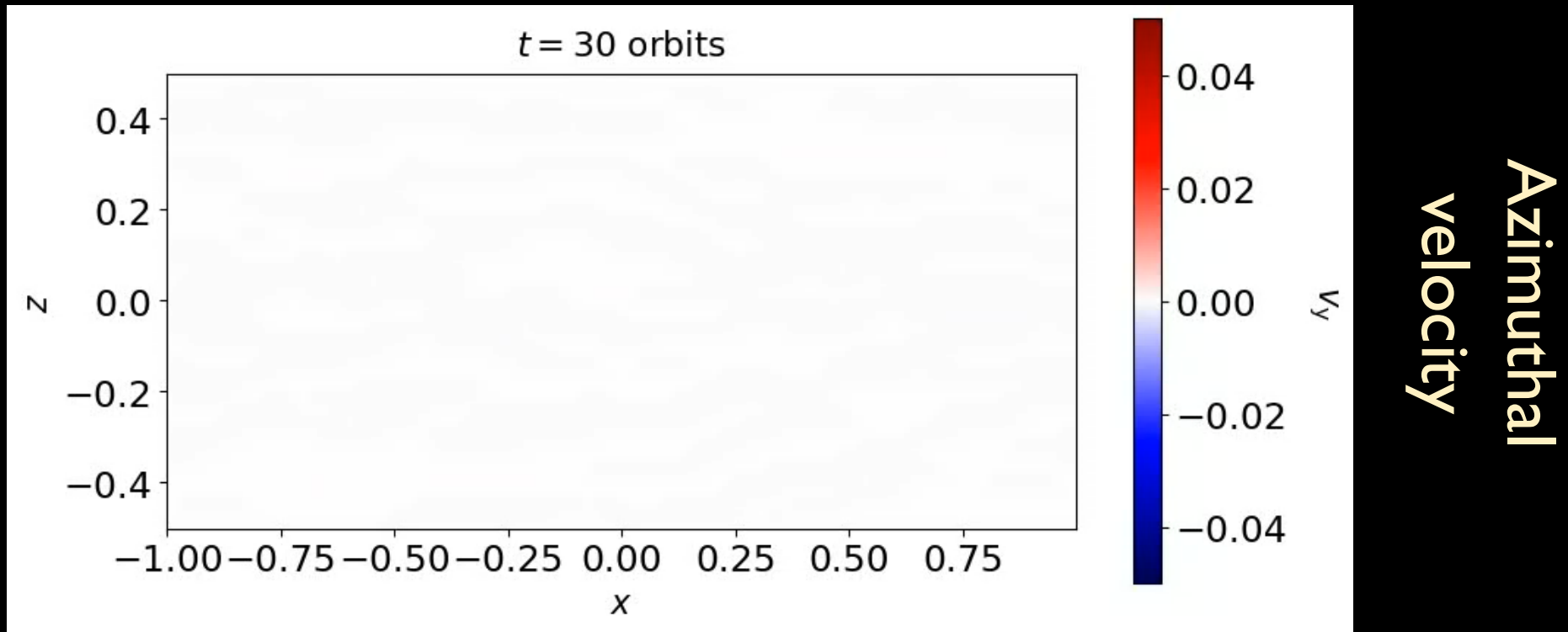
R-DDI happens if
 $0.02 < \Lambda \lesssim 0.10$

MRI excluded if
 $k_{MRI} < 2\pi/H$

What does
the R-DDI look like?

It forms zonal flows
like the COS!

R-DDI forms zonal flows! (*pressure bumps!*)



$$\Lambda = 1$$

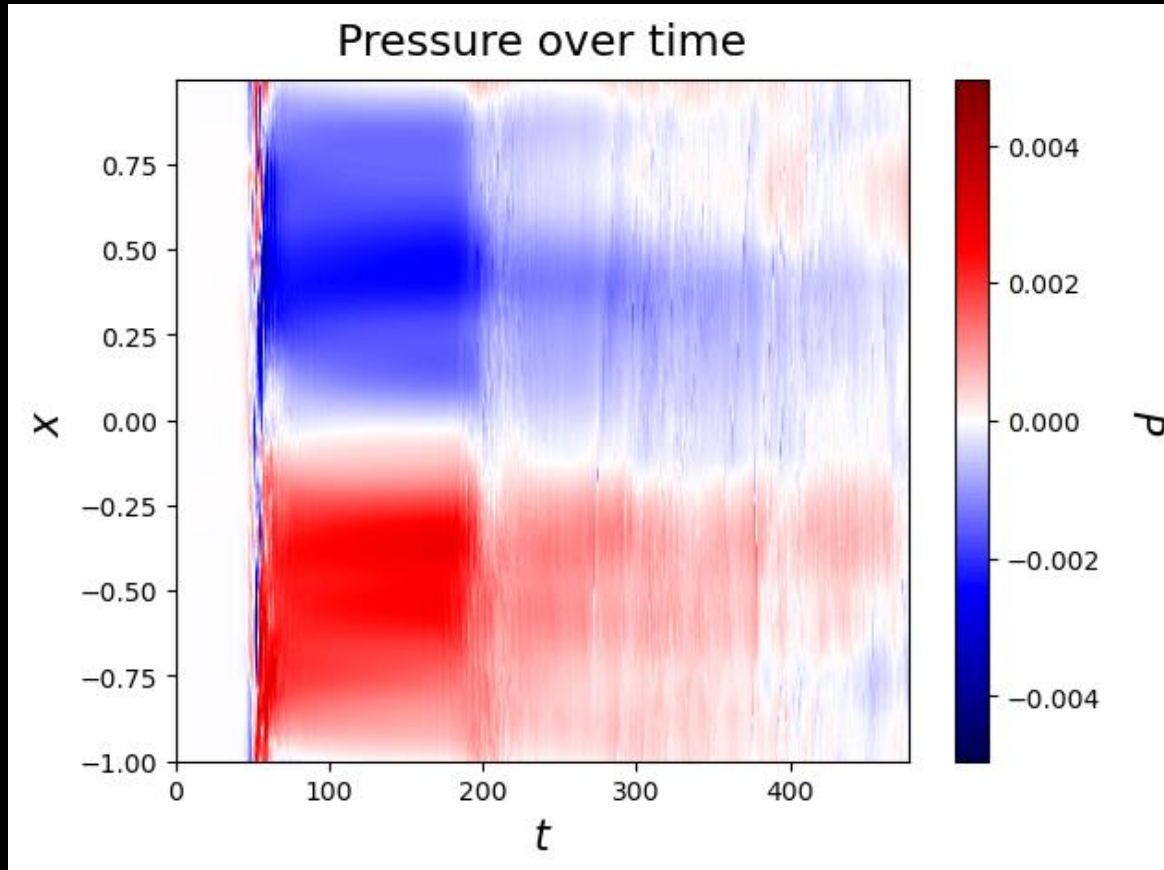
$$q = 10^{-6}$$

$$Re = 10^5$$

super- v_K sub- v_K

Spectral hydro simulation with
dedalus

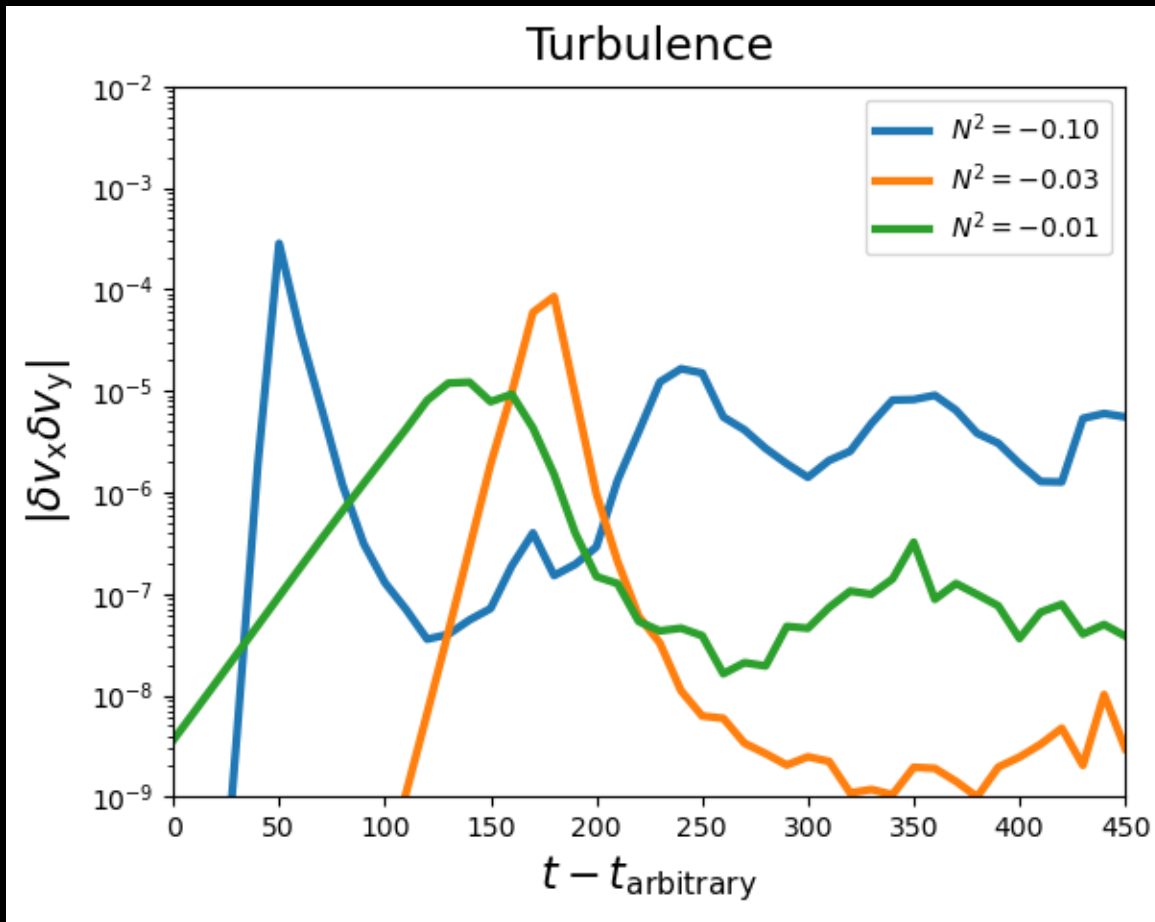
R-DDI forms pressure bumps!



$$\Lambda = 1$$
$$q = 10^{-6}$$
$$Re = 10^5$$

In 3-D, pressure bumps would probably turn into vortices (like the COS).

Turbulence induced by R-DDI is weak

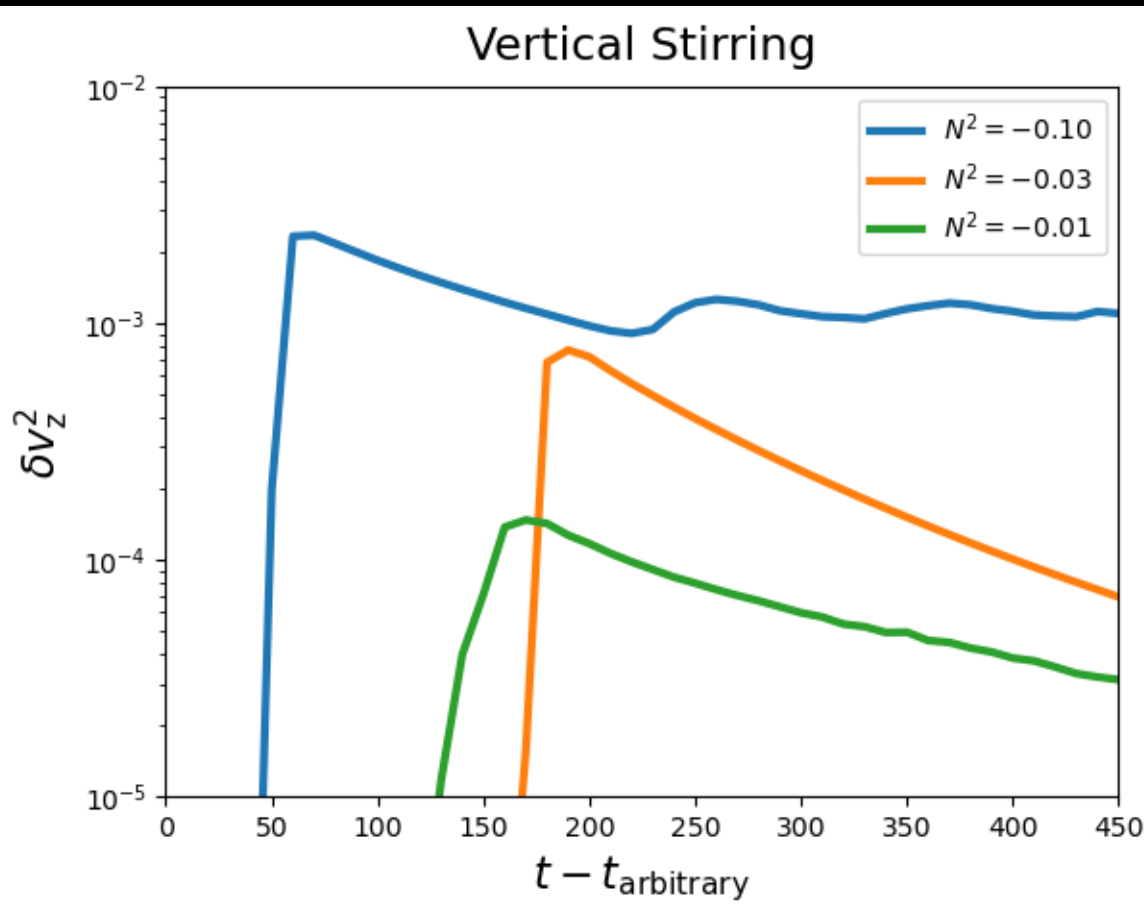


Induces low levels
of turbulence

$$\alpha_{\text{peak}} \approx 10^{-4}$$

Turbulence also weakens after peak!

Vertical stirring by R-DDI is stronger



Induces stronger
vertical stirring
than turbulence

$$\alpha_{z,\text{peak}} \approx 10^{-3}$$

R-DDI with a strong N^2 could
potentially inhibit the streaming instability.

Summary

The convective overstability (COS)
can survive magnetic fields if $\Lambda \leq 10^{-2}$.

The resistive double-diffusive instability (R-DDI)
can happen at $10^{-2} \leq \Lambda \leq 10^{-1}$ provided $q < 10^{-2}$.

Like the COS, the R-DDI produces zonal flows
that can help trap dust.

(good for planetesimal formation!)

But strong R-DDI can also generate enough vertical
stirring of dust to inhibit planetesimal formation.